]

SOFTWARE REQUIREMENTS SPECIFICATION

Hotel Automation Software

Document Type	Software Requirements Specification (SRS)
Subject	Software Engineering
Subject Code	CS3004
Date	April 22, 2026
Group Members	Suyash Pandey : 2023UG1100 Suraj Kumar : 2023UG1106 Himanshu Raj : 2023UG1092 Satyendra Kumar : 2023UG1105
Institution	Indian Institute of Information Technology Ranchi, Jharkhand

Contents

1 Introduction **3**
1.1 Purpose . . . 3
1.2 Scope . . . 3
1.3 Definitions, Acronyms, and Abbreviations . . . 3
1.4 Overview . . . 4

2 Overall Description **5**
2.1 Product Perspective . . . 5
2.2 Product Functions . . . 5
2.3 User Classes and Characteristics . . . 5
2.4 Operating Environment . . . 6
2.5 Design and Implementation Constraints . . . 6
2.6 Assumptions and Dependencies . . . 7

3 Specific Functional Requirements **8**
3.1 Room Management Requirements . . . 8
3.1.1 Room Inventory Configuration . . . 8
3.1.2 Occupancy Reporting and Tariff Revision . . . 8
3.2 Reservation Requirements . . . 8
3.2.1 Guest Data Entry . . . 8
3.2.2 Room Allocation . . . 9
3.3 Catering Services Requirements . . . 9
3.4 Billing and Check-Out Requirements . . . 9
3.5 Frequent-Guest Loyalty Requirements . . . 10
3.6 User Interface Requirements . . . 10

4 Non-Functional Requirements **12**
4.1 Performance Requirements . . . 12
4.2 Usability Requirements . . . 12
4.3 Reliability Requirements . . . 12
4.4 Maintainability Requirements . . . 13
4.5 Portability Requirements . . . 13
4.6 Security Requirements . . . 13

5 Appendix **14**
5.1 Use-Case Summary . . . 14
5.2 Data Dictionary . . . 14
5.3 Requirements Traceability Matrix . . . 15
5.4 Revision History . . . 16

1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the **Hotel Automation Software**. The purpose of this document is to provide a comprehensive, unambiguous description of the system's behaviour to guide development, testing, and validation of the software.

This software will automate the day-to-day operational activities of a 5-star hotel, including room management, guest reservation, catering services, billing, and loyalty programme management. The SRS serves as a contractual agreement between stakeholders and the development team.

1.2 Scope

The Hotel Automation Software is designed to manage the complete operational workflow of a 5-star hotel, from room allocation and guest check-in through to catering, billing, and check-out. The software will:

- Maintain a database of all hotel rooms categorised by bed type (single / double) and amenity type (AC / Non-AC), along with their associated tariffs.
- Allow dynamic revision of room tariffs based on the prevailing occupancy rate, adjusted by a configurable percentage either upward or downward.
- Support advance and walk-in room reservations, subject to real-time room availability.
- Assign unique room numbers and token numbers to accommodated guests and generate apology messages when accommodation is unavailable.
- Record catering orders (food item, quantity, guest token, date and time) entered by the catering services manager.
- Generate itemised bills at check-out, including room charges and catering charges, and print the net balance payable.
- Maintain a frequent-guest identity database and apply applicable discounts to bills.

The software is **NOT** intended to:

- Control physical room infrastructure (lighting, HVAC, locks) in real time.
- Handle payroll or HR management for hotel staff.
- Provide travel or tourism booking services beyond hotel accommodation.

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification

Term / Acronym	Definition
AC	Air-Conditioned — denotes a room equipped with air conditioning
Non-AC	Non-Air-Conditioned — denotes a room without air conditioning
Room Tariff	The per-night charge for a specific room, which may vary by room type (single/double, AC/Non-AC) and by season / occupancy rate
Occupancy Rate	The percentage of hotel rooms that are occupied at a given point in time; used by management to decide tariff adjustments
Token Number	A unique identifier assigned to each guest at check-in; used to associate catering orders and other services with the correct guest
Advance Reservation	A room booking made prior to the guest's arrival date
Walk-in Reservation	A room booking made by a guest who arrives at the hotel without a prior booking
Receptionist	Hotel staff member responsible for recording guest data, assigning rooms, and issuing token numbers
Catering Services Manager	Hotel staff member responsible for recording food and beverage orders against guest token numbers
Frequent Guest	A returning guest who has been issued a loyalty identity number and is eligible for special discounts
Identity Number	A unique loyalty identifier issued to frequent guests that entitles them to discount benefits
GUI	Graphical User Interface
RDBMS	Relational Database Management System — used to persist room, guest, catering, and billing data
SA/SD	Structured Analysis / Structured Design
UML	Unified Modelling Language

1.4 Overview

The remainder of this document is organised as follows:

- **Section 2** provides the overall description of the system, including product perspective, functions, users, constraints, and assumptions.
- **Section 3** details the specific functional requirements of the system.
- **Section 4** details the non-functional requirements including performance, usability, reliability, and portability.
- **Section 5** contains the appendix with a use-case summary, a data dictionary, a traceability matrix, and the revision history.

2 Overall Description

2.1 Product Perspective

The Hotel Automation Software is a standalone desktop application implemented in **Java** with a **Java Swing**-based graphical user interface, backed by a relational database for persistent storage of rooms, guests, catering records, and billing data. It does not interface with any external reservation portal or payment gateway in this version.

The system interacts with the following entities:

- **Hotel Manager:** Configures room types, tariffs, and discount policies; reviews occupancy reports and adjusts tariffs.
- **Receptionist:** Records guest arrival data, processes reservations, assigns rooms and token numbers, and initiates check-out billing.
- **Catering Services Manager:** Enters food and beverage orders against guest token numbers during the guest's stay.
- **Guest:** The end beneficiary; interacts with the system indirectly through the receptionist and catering staff.

2.2 Product Functions

At a high level, the system shall:

1. Maintain a configurable inventory of rooms classified by bed type and AC type, each with an associated tariff.
2. Display the average occupancy rate for any given month to support management tariff decisions.
3. Allow the manager to revise room tariffs upward or downward by a specified percentage.
4. Accept guest reservation requests (advance or walk-in) with arrival date, advance payment, approximate duration of stay, and room type preference.
5. Allocate an available room and assign a unique token number to each accommodated guest; generate an apology message when no suitable room is available.
6. Record catering orders (item, quantity, token number, date and time) entered by the catering services manager.
7. Generate a complete itemised bill at check-out comprising room charges and catering charges, and display the net balance payable after deducting any advance payment.
8. Issue loyalty identity numbers to frequent guests and apply the applicable discount to their bills.

2.3 User Classes and Characteristics

User Class	Description	Privilege Level
Hotel Manager	Configures the system (rooms, tariffs, discounts) and reviews occupancy and revenue reports.	Administrative access
Receptionist	Handles guest reservations, room assignments, token issuance, and check-out processing.	Operational access
Catering Services Manager	Records food and beverage orders against guest token numbers.	Catering module access
System Administrator	Installs and maintains the application and database on hotel workstations.	Configuration and maintenance access

2.4 Operating Environment

- **Platform:** Desktop application (Windows 10+ / Ubuntu 20.04+ / macOS 11+)
- **Implementation Language:** Java (JDK 8 or above)
- **User Interface:** Java Swing
- **Database:** Embedded relational database (e.g., SQLite or Apache Derby)
- **Minimum RAM:** 512 MB
- **Minimum Disk Space:** 200 MB (including database storage)
- **Display:** Minimum 1024 × 768 resolution
- **Reference Hardware for Performance Targets:** Intel Core i5 (8th Gen or equivalent), 8 GB RAM, SSD storage

2.5 Design and Implementation Constraints

- The system must be implemented entirely in Java.
- The user interface must be developed using Java Swing.
- All persistent data (rooms, guests, catering records, bills) must be stored in a relational database; no flat-file storage shall be used for transactional data.
- The application shall be delivered as a single executable JAR file bundling the embedded database driver.
- No external payment gateway or network connectivity is required in this version.
- Token numbers must be unique across the lifetime of the system, not merely within a single day.

2.6 Assumptions and Dependencies

- Room tariffs are defined per room type (bed type × AC type) and are uniform within a type; individual room-level pricing is not supported in this version.
- Tariff adjustments apply to all rooms of the targeted type simultaneously.
- A guest is uniquely identified by their token number for the duration of their stay; the same guest may receive a new token number on each visit unless recognised as a frequent guest.
- Catering charges are computed from the quantities and unit prices stored in the system at the time of bill generation.
- The system assumes inputs are provided in good faith; basic input validation is required but adversarial inputs are not a primary concern.
- The "approximate duration of stay" entered at check-in is for planning purposes only; actual charges are computed from the real check-out date and time.

3 Specific Functional Requirements

3.1 Room Management Requirements

3.1.1 Room Inventory Configuration

FR-1.1: The system shall allow the hotel manager to define and maintain a list of rooms, each characterised by a unique room number, bed type (single or double), and AC type (AC or Non-AC).

FR-1.2: The system shall associate a base tariff (per night, in currency units) with each combination of bed type and AC type (i.e., up to four tariff categories: single-AC, single-Non-AC, double-AC, double-Non-AC).

FR-1.3: The system shall track the current availability status of each room (*available* or *occupied*).

3.1.2 Occupancy Reporting and Tariff Revision

FR-2.1: The system shall compute and display the average occupancy rate for any given month, defined as:

$$\text{Average Occupancy Rate (\%)} = \frac{\text{Total Room-Nights Occupied}}{\text{Total Room-Nights Available}} \times 100$$

FR-2.2: The system shall allow the hotel manager to revise the tariff for any room type by specifying a percentage adjustment (positive for upward revision, negative for downward revision). The new tariff shall be computed as:

$$\text{New Tariff} = \text{Current Tariff} \times \left(1 + \frac{\text{Adjustment \%}}{100}\right)$$

FR-2.3: The system shall store a history of tariff changes (room type, old tariff, new tariff, date of change) to support audit and reporting.

3.2 Reservation Requirements

3.2.1 Guest Data Entry

FR-3.1: The system shall allow the receptionist to enter the following data for each reservation request:

- Guest name
- Arrival date and time
- Approximate duration of stay (in nights)
- Room type required (bed type and AC type)
- Advance payment amount (zero for walk-in guests with no advance)

- Frequent-guest identity number (optional; leave blank if not a frequent guest)

FR-3.2: The system shall support both advance reservations (made before the arrival date) and walk-in reservations (made on the day of arrival).

3.2.2 Room Allocation

FR-4.1: Upon receiving a reservation request, the system shall check the availability of rooms matching the requested type on the requested dates.

FR-4.2: If a suitable room is available, the system shall:

- Allocate the room with the lowest available room number of the requested type.
- Mark the room as *occupied* for the duration of the stay.
- Assign a unique token number to the guest.
- Display and print the allocated room number and token number to the receptionist.

FR-4.3: If no suitable room is available, the system shall generate and display an apology message to the receptionist indicating the unavailability and the reason (no room of the requested type available on the requested dates).

FR-4.4: The system shall not double-allocate a room to two overlapping reservations.

3.3 Catering Services Requirements

FR-5.1: The system shall allow the catering services manager to enter catering orders comprising:

- Guest token number
- Food item name
- Quantity consumed
- Date and time of consumption

FR-5.2: The system shall validate that the token number entered corresponds to a currently checked-in guest, and shall display an error message if not.

FR-5.3: The system shall maintain a menu of food items with associated unit prices, configurable by the hotel manager.

FR-5.4: The system shall accumulate all catering orders for a guest throughout their stay and associate them with the guest's token number for billing purposes.

3.4 Billing and Check-Out Requirements

FR-6.1: When a guest prepares to check out, the receptionist shall initiate the check-out process by entering the guest's token number.

FR-6.2: The system shall compute the total room charge as:

$$\text{Room Charge} = \text{Tariff per Night (at time of check-in)} \times \text{Actual Nights Stayed}$$

FR-6.3: The system shall compute the total catering charge as the sum of (quantity × unit price) across all catering orders associated with the guest's token number.

FR-6.4: The system shall generate and display an itemised bill comprising:

- Guest name, room number, token number, check-in and check-out dates
- Itemised room charges (tariff per night × nights)
- Itemised catering charges (food item, quantity, unit price, subtotal per item)
- Total bill amount (room + catering)
- Advance payment deducted
- Frequent-guest discount deducted (if applicable)
- Net balance payable

FR-6.5: The system shall print the bill to the system's default printer upon receptionist confirmation.

FR-6.6: Upon successful check-out, the system shall mark the room as *available* from the check-out date.

3.5 Frequent-Guest Loyalty Requirements

FR-7.1: The system shall maintain a frequent-guest registry storing the guest's name, contact information, and a unique system-generated identity number.

FR-7.2: The hotel manager shall be able to define the discount percentage applicable to frequent guests. The discount shall apply to the total bill (room + catering charges) before deducting advance payments.

FR-7.3: When a reservation is made using a valid frequent-guest identity number, the system shall automatically apply the configured discount at check-out.

FR-7.4: The system shall allow the receptionist to enrol a new frequent guest during the check-out process, issuing a new identity number that is effective from the next stay.

3.6 User Interface Requirements

FR-8.1: The system shall provide a graphical user interface built using Java Swing, with separate functional panels for Room Management, Reservations, Catering, Billing, and Reports.

FR-8.2: The GUI shall provide input forms for all reservation parameters described in FR-3.1, with field-level validation and descriptive error messages.

FR-8.3: The GUI shall provide a **Check Availability** button that queries room availability before committing a reservation.

FR-8.4: The GUI shall provide a **Confirm Reservation** button to finalise the allocation and issue the token number.

FR-8.5: The GUI shall provide a **Check-Out** button that initiates bill generation for the guest identified by the entered token number.

FR-8.6: The GUI shall provide a **Print Bill** button that sends the generated bill to the system's default printer.

FR-8.7: The GUI shall provide an **Adjust Tariff** panel accessible to the hotel manager, allowing percentage-based tariff revisions per room type.

FR-8.8: The GUI shall provide an **Occupancy Report** panel displaying the average occupancy rate for a user-selected month.

FR-8.9: The system shall validate all user inputs before processing and display specific, actionable error messages for each invalid input field.

FR-8.10: All input fields shall carry descriptive labels and tooltips explaining expected values, formats, and valid ranges.

4 Non-Functional Requirements

4.1 Performance Requirements

ID	Requirement	Description	Metric
NFR-P1	Query Response	Room availability queries and occupancy reports shall return results promptly on the reference hardware (Intel Core i5 8th Gen, 8 GB RAM) for a hotel with up to 500 rooms and 5 years of historical data.	$<$ 2 seconds
NFR-P2	Bill Generation	Generating and displaying the itemised check-out bill shall complete quickly, regardless of the number of catering line items.	$<$ 3 seconds
NFR-P3	UI Responsiveness	The GUI shall remain responsive during all database operations (operations execute on a background thread).	No EDT freeze $>$ 500 ms
NFR-P4	Startup Time	The application shall start and be ready for user input promptly after launch, including database initialisation.	$<$ 5 seconds

4.2 Usability Requirements

- **NFR-U1:** The system shall provide a user interface that requires no programming knowledge to operate.
- **NFR-U2:** All input fields shall have descriptive labels and tooltips explaining expected values, formats, and valid ranges (see also FR-8.10).
- **NFR-U3:** Error messages shall be specific, identifying the offending field and the corrective action required.
- **NFR-U4:** Billing output shall be presented in a clearly formatted, itemised layout readable without domain expertise.
- **NFR-U5:** A first-time receptionist with basic computer skills shall be able to complete a reservation and check-out within 10 minutes without consulting documentation.

4.3 Reliability Requirements

- **NFR-R1:** The system shall not crash or produce erroneous results for any valid input combination.

- **NFR-R2:** The system shall handle all boundary conditions gracefully, including zero available rooms, a guest with no catering orders, and a stay of zero nights.
- **NFR-R3:** No two guests shall be allocated the same room for any overlapping date range.
- **NFR-R4:** All financial calculations (tariff application, discount, net balance) shall be accurate to two decimal places.
- **NFR-R5:** The system shall preserve all reservation and billing records across application restarts; no transactional data shall be lost due to an unexpected shutdown after the transaction has been committed to the database.

4.4 Maintainability Requirements

- **NFR-M1:** The source code shall follow standard Java coding conventions (Oracle Code Conventions) and every public class and method shall have a Javadoc comment.
- **NFR-M2:** The codebase shall be structured in a modular manner with clear separation between: database access layer, business logic, and UI components.
- **NFR-M3:** Adding a new room type or a new food item shall require only data-level changes (inserts into the room-type or menu table) with no modification to business logic code.

4.5 Portability Requirements

- **NFR-PO1:** The software shall run on any platform supporting Java Runtime Environment (JRE 8 or above), including Windows 10+, Ubuntu 20.04+, and macOS 11+.
- **NFR-PO2:** The application shall be packaged as a single executable JAR file bundling the embedded database driver, requiring no external installation.
- **NFR-PO3:** The software shall not depend on any operating-system-specific libraries or absolute file paths.

4.6 Security Requirements

- **NFR-S1:** The application shall support role-based access control, ensuring that catering staff cannot access billing or tariff management functions, and receptionists cannot access system configuration.
- **NFR-S2:** Exported or printed bills shall contain only guest transaction data and shall not include any system-level information (OS version, database paths, internal IDs).
- **NFR-S3:** Input validation (FR-8.9) shall prevent the application from entering an undefined or erroneous state due to malformed, empty, or out-of-range inputs.
- **NFR-S4:** Guest personal data (name, contact details) shall be stored only in the local database and shall not be transmitted to any external system.

5 Appendix

5.1 Use-Case Summary

UC ID	Use Case	Actor(s)	Brief Description
UC-01	Make Reservation	Receptionist	Guest requests a room; receptionist enters details; system allocates room and issues token.
UC-02	Handle No Availability	Receptionist	No suitable room found; system generates apology message.
UC-03	Record Catering Order	Catering Mgr	Catering manager enters food items against a guest token number.
UC-04	Process Check-Out	Receptionist	Receptionist initiates check-out; system generates and prints itemised bill.
UC-05	Adjust Room Tariff	Hotel Manager	Manager reviews occupancy report and revises tariff by a percentage.
UC-06	View Occupancy Report	Hotel Manager	Manager selects a month; system displays average occupancy rate.
UC-07	Enrol Frequent Guest	Receptionist	Receptionist enrols a qualifying guest; system issues a loyalty identity number.
UC-08	Apply Loyalty Discount	Receptionist	Frequent guest's identity number is entered; system applies configured discount at check-out.

5.2 Data Dictionary

Attribute	Data Type	Constraint	Description
`room_number`	Integer	PK, ≥ 1	Unique identifier for each physical room
`bed_type`	Enum	{Single, Double}	Type of bed in the room
`ac_type`	Enum	{AC, Non-AC}	Air-conditioning category of the room
`tariff_per_night`	Decimal	> 0	Current nightly rate for the room type
`token_number`	Integer	PK, unique, auto	System-generated unique guest token

Attribute	Data Type	Constraint	Description
`guest_name`	String	Non-empty	Full name of the guest
`arrival_datetime`	DateTime	Not null	Check-in date and time
`checkout_datetime`	DateTime	$\geq$ arrival	Actual check-out date and time
`advance_paid`	Decimal	≥ 0	Amount paid in advance by the guest
`identity_number`	Integer	FK (optional)	Frequent-guest loyalty ID; null if not enrolled
`food_item`	String	Non-empty	Name of the food or beverage item ordered
`quantity`	Integer	≥ 1	Number of units of the food item consumed
`unit_price`	Decimal	> 0	Price per unit of the food item
`order_datetime`	DateTime	Not null	Date and time the catering order was placed
`discount_pct`	Decimal	$0 \leq d \leq 100$	Percentage discount applied to frequent guests

5.3 Requirements Traceability Matrix

Req. ID	Requirement Description	Priority	Module	Status
FR-1.1 – FR-1.3	Room inventory, type classification, availability tracking	**High**	Room Mgmt	Pending
FR-2.1 – FR-2.3	Occupancy reporting, tariff revision, change history	**High**	Room Mgmt	Pending
FR-3.1 – FR-3.2	Guest data entry, advance and walk-in reservation support	**High**	Reservation	Pending
FR-4.1 – FR-4.4	Room availability check, allocation, token issuance, apology message	**High**	Reservation	Pending
FR-5.1 – FR-5.4	Catering order entry, validation, menu management, accumulation	**High**	Catering	Pending
FR-6.1 – FR-6.6	Check-out initiation, room/catering charge computation, itemised bill, print, room release	**High**	Billing	Pending
FR-7.1 – FR-7.4	Frequent-guest registry, discount config, discount application, enrolment	**Medium**	Loyalty	Pending

Req. ID	Requirement Description	Priority	Module	Status
FR-8.1 – FR-8.10	Java Swing GUI, reservation/catering/billing panels, validation	**High**	UI Layer	Pending
NFR-R3	No double-allocation of rooms for overlapping dates	**High**	Reservation	Pending
NFR-R4	Financial calculations accurate to two decimal places	**High**	Billing	Pending
NFR-S1	Role-based access control for staff user classes	**Medium**	UI Layer	Pending

5.4 Revision History

Version	Date	Author	Description
0.1	April 2026	Project Team	Initial Draft
1.0	April 22, 2026	Project Team	First complete draft; covers room management, reservation, catering, billing, and frequent-guest requirements; non-functional requirements; use-case summary; data dictionary; traceability matrix

End of Document — Hotel Automation Software SRS v1.0